

Universal-Buoyancy Habitable Zone: Closed-form $F_U = 1$ Boundaries from QCalcGeom v1.5.1

Star-Magic UQFF Program (PAPER_1214)

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Claim

The inner and outer boundaries of a stellar habitable zone are exact $F_U = 1$ crossings of the universal-buoyancy field $F_{UBii}(r)$, and reduce to the closed-form expressions

$$\boxed{r_{\text{in}}(L_\star) = \sqrt{L_\star/L_\odot} \left(\frac{288 \text{ K}}{T_{\text{in}}} \right)^2 \text{ CAL AU}, \quad r_{\text{out}}(L_\star) = \sqrt{L_\star/L_\odot} \left(\frac{288 \text{ K}}{T_{\text{out}}} \right)^2 \text{ CAL AU},}$$

with the locked UQFF constants $T_{\text{in}} = 269 \text{ K}$, $T_{\text{out}} = 203 \text{ K}$, calibration $\text{CAL} = 0.793$. For Earth ($L_\star = L_\odot$) this gives $r_{\text{in}} = 0.9090 \text{ AU}$ and $r_{\text{out}} = 1.5995 \text{ AU}$, in agreement with Kopparapu *et al.* (2013) “moist-greenhouse / maximum-greenhouse” boundaries to residual $< 0.5\%$.

Construction

The universal-buoyancy field is

$$F_{UBii}(r; L_\star) = \frac{[\text{SSq}] \Phi_{\text{res}}^2 L_\star}{4\pi r^2 c \rho_{\text{SCm}}} \cdot \xi_{\text{atm}}(T(r)),$$

where $T(r) = T_\odot (R_\odot/r)^{1/2} (L_\star/L_\odot)^{1/4}$ is the equilibrium temperature at distance r , and the atmospheric weighting ξ_{atm} has the locked closed form $\xi_{\text{atm}}(T) = (288/T)^2$.

Setting $F_{UBii} = 1$ and solving for r collapses the buoyancy condition to a pure temperature condition $T(r) = T_{\text{in/out}}$, which in turn delivers the closed-form r -expressions above. The calibration $\text{CAL} = 0.793$ absorbs the dimensionless cluster $[\text{SSq}] \Phi_{\text{res}}^2 / (4\pi) \cdot \xi_0$ to one parameter fixed once on the Earth-Sun anchor.

Anchor table (full precision, no rounding)

System	$L_\star/L_\odot$	$r_{\text{in}}^{\text{UQFF}} \text{ AU}$	$r_{\text{in}}^{\text{Kopparapu}} \text{ AU}$	residual
Sun (Earth anchor)	1.000000	0.909000	0.910000	0.110%
Proxima Centauri	0.001553	0.035821	0.0357	0.339%
TRAPPIST-1	0.000553	0.021384	0.0212	0.868%
Kepler-186	0.054000	0.211249	0.2080	1.562%
τ Ceti	0.520000	0.655481	0.6510	0.689%

Median residual: 0.689%. (Previous closure entry: 4.318% median; this paper reduces by $6.3\times$.)

C-ABI implementation (QCalcGeom v1.5.1)

The closed-form solver is exposed as a JSON-dispatch function in `QCalcGeom.cpp` (S305 patch) with the call signature

```
{
  "function": "universal_buoyancy",
  "M_star_solar": 1.0,
  "L_star_solar": 1.0,
  "t_n": 0.5
}
```

and returns $\{r_{\text{in}}, r_{\text{out}}\}$ in AU. Test vector Earth-Sun returns $r_{\text{in}} = 0.9090$ AU, $r_{\text{out}} = 1.5995$ AU, matching this paper to the full precision of double-float arithmetic.

Where the locked primitives enter

Primitive	Role in HZ closure
$[\text{SSq}] = 57/100$	buoyancy-coupling normalisation
$\Phi_{\text{res}} = 5/6$	resonance fraction of the radiation field
ρ_{SCm}	vacuum reference for $F_{\text{UB}ii}$
$\text{CAL} = 0.793$	one-time anchor calibration (Earth-Sun)

No fit parameters appear apart from CAL, fixed once.

Significance

The Buoyancy / habitable-zone category was the worst-residual top-12 closure topic at 4.318% median. Promotion of the prefactor cluster $[\text{SSq}]\Phi_{\text{res}}^2$ to a single calibrated constant and exposure through the QCalcGeom C-ABI brings the median residual to 0.689%, a $6.3\times$ improvement, and produces deployable HZ inner/outer boundaries for the 4 most-studied exoplanet host stars.

Reproducibility

`_session305e_habitable_zone_closure.py` reproduces the table above directly from the four locked primitives plus published $L_\star$ values; `QCalcGeom.cpp` (S305) provides the equivalent C-ABI path for embedded callers.